Boss Challenge — Paper 09 (Class 7)

Rational Numbers | Motion & Time | Data Handling | Heat

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A number is called rational only if it can be written as p/q where p and q are integers and:

(A) p is zero
(B) q is zero
(C) p equals q
(D) q is not zero

2. The speed of a moving object is found by:

(A) multiplying distance by time
(B) dividing the time taken by the distance
(C) adding distance and time
(D) dividing the distance travelled by the time taken

3. The arithmetic mean (average) of a set of numbers is found by:

(A) choosing the value that appears most often
(B) picking the middle value
(C) subtracting the smallest from the largest
(D) adding all the values and dividing by how many there are

4. Heat always flows on its own from a body at a ____ temperature to a body at a ____ temperature.

(A) equal; equal
(B) lower; higher
(C) higher; higher
(D) higher; lower

5. Why is every integer also a rational number?

(A) because any integer can be written as itself over 1
(B) because integers have no sign
(C) because integers cannot be written as fractions
(D) because integers are always positive

6. The basic (SI) unit of speed is:

(A) metre per second squared (m/s^2)
(B) second (s)
(C) kilometre (km)
(D) metre per second (m/s)

7. Find the mean of 4, 6, 8 and 10.

(A) 8
(B) 28
(C) 6
(D) 7

8. The instrument used to measure temperature is a:

- **(A)** thermometer
- **(B)** ruler
- **(C)** speedometer
- **(D)** barometer

9. The rational number -3/4 lies between which two integers on the number line?

- **(A)** -3 and -2
- **(B)** -4 and -3
- **(C)** -1 and 0
- **(D)** 0 and 1

10. Find the speed of a car that travels 60 km in 2 hours.

- **(A)** 120 km/h
- **(B)** 30 km/h
- **(C)** 62 km/h
- **(D)** 15 km/h

11. The mode of a set of data is:

- **(A)** the value that occurs most often
- **(B)** the largest value
- **(C)** the average of all values
- **(D)** the middle value

12. A healthy human body normally stays at a temperature of about:

- **(A)** 50 degC
- **(B)** 37 degC
- **(C)** 0 degC
- **(D)** 100 degC

13. Express 18/24 in its lowest terms.

- **(A)** 6/8
- **(B)** 3/4
- **(C)** 9/12
- **(D)** 2/3

14. A bus covers 150 m in 10 seconds. Its speed is:

- **(A)** 10 m/s
- **(B)** 15 m/s
- **(C)** 160 m/s
- **(D)** 1500 m/s

15. Find the mode of 2, 3, 3, 5, 7, 3 and 9.

- **(A)** 7
- **(B)** 5
- **(C)** 3
- **(D)** 9

16. A clinical thermometer is built to read temperatures roughly in the range:

(A) 0 degC to 100 degC
(B) -10 degC to 10 degC
(C) 100 degC to 200 degC
(D) 35 degC to 42 degC

17. Which rational number is greater: -2/3 or -3/4?

(A) -2/3
(B) they cannot be compared
(C) -3/4
(D) they are equal

18. An object moving at constant (uniform) speed covers:

(A) different distances each second
(B) equal distances in equal intervals of time
(C) less distance in each later second
(D) more distance in each later second

19. The median of a set of data is:

(A) the most frequent value
(B) the sum of all values
(C) the difference of the largest and smallest
(D) the middle value when the data are arranged in order

20. Heat travels through a solid metal rod mainly by the process of:

(A) radiation
(B) evaporation
(C) conduction
(D) convection

21. Find -3/5 + 1/5.

(A) 2/5
(B) -2/10
(C) -4/5
(D) -2/5

22. A pendulum bob's single complete to-and-fro swing is called one:

(A) oscillation
(B) time period
(C) amplitude
(D) frequency

23. Find the median of 5, 2, 9, 7 and 3.

(A) 5
(B) 7
(C) 2
(D) 9

24. Heat is transferred through liquids and gases mainly by:

(A) convection
(B) conduction
(C) radiation
(D) freezing

25. Add the fractions 1/2 and (-1/3).

(A) 2/5
(B) 1/5
(C) -1/6
(D) 1/6

26. The time taken for one complete oscillation of a pendulum is its:

(A) speed
(B) time period
(C) amplitude
(D) frequency

27. Find the median of 4, 8, 6 and 10.

(A) 9
(B) 8
(C) 7
(D) 6

28. Heat from the Sun reaches the Earth across empty space by:

(A) radiation
(B) conduction
(C) only through the air
(D) convection

29. Subtract -2/7 from 3/7.

(A) -5/7
(B) 5/7
(C) 5/14
(D) 1/7

30. A pendulum completes 20 oscillations in 40 seconds. Its time period is:

(A) 20 s
(B) 2 s
(C) 40 s
(D) 800 s

31. The range of the data 12, 7, 9, 20 and 4 is:

(A) 24
(B) 16
(C) 4
(D) 20

32. Among these materials, which is the BEST conductor of heat?

(A) copper
(B) plastic
(C) air
(D) wood

33. Find (-2/3) x (3/5).

(A) -2/5
(B) 2/5
(C) -5/6
(D) -6/8

34. The speedometer of a vehicle measures its:

(A) total distance travelled
(B) speed
(C) fuel left in the tank
(D) time of the journey

35. In a bar graph, the size of each value is shown by the ____ of its bar.

(A) position only
(B) width
(C) height (length)
(D) colour

36. Materials that do NOT let heat pass through them easily are called:

(A) radiators
(B) conductors
(C) insulators
(D) metals

37. Find (-4/9) divided by (2/3).

(A) -6/11
(B) -2/3
(C) 2/3
(D) -8/27

38. The odometer of a vehicle measures the:

(A) speed of the vehicle
(B) engine temperature
(C) time taken
(D) total distance travelled

39. A double bar graph is most useful when you want to:

(A) show a single set of data
(B) measure time only
(C) hide the larger values
(D) compare two sets of data side by side

40. In winter we prefer to wear woollen clothes mainly because wool:

- **(A)** traps air and is a poor conductor, keeping body heat in
- **(B)** is a very good conductor of heat
- **(C)** produces heat by itself
- **(D)** reflects all the cold into the body

41. The reciprocal of -5/7 is:

- **(A)** -7/5
- **(B)** 5/7
- **(C)** -5/7
- **(D)** 7/5

42. On a distance-time graph, a straight line sloping upward shows the object is moving with:

- **(A)** steadily increasing speed
- **(B)** constant (uniform) speed
- **(C)** backward motion
- **(D)** zero speed

43. The maximum daily temperatures for a week were 30, 31, 32, 30, 31, 32, 31 (in degC). Their mean is:

- **(A)** 30 degC
- **(B)** 31 degC
- **(C)** 32 degC
- **(D)** 33 degC

44. During the daytime, a sea breeze blows from the:

- **(A)** land towards the sea
- **(B)** ground towards the sky only
- **(C)** east towards west always
- **(D)** sea towards the land

45. The additive inverse of -4/9 is:

- **(A)** 9/4
- **(B)** 4/9
- **(C)** 0
- **(D)** -4/9

46. On a distance-time graph, a horizontal (flat) straight line means the object is:

- **(A)** at rest
- **(B)** moving very fast
- **(C)** slowing down
- **(D)** speeding up

47. The chance (probability) of getting a head when tossing a fair coin is:

(A) 1 out of 2
(B) impossible
(C) 1 out of 6
(D) certain to happen

48. The handle of a good cooking pan is made of plastic or wood because these materials are:

(A) poor conductors (insulators) of heat
(B) excellent conductors of heat
(C) able to produce heat
(D) magnetic materials

49. A rational number lying between 1/4 and 1/2 is:

(A) 5/8
(B) 3/4
(C) 1/8
(D) 3/8

50. Convert a speed of 72 km/h into metres per second.

(A) 2 m/s
(B) 200 m/s
(C) 20 m/s
(D) 72 m/s

51. When you roll an ordinary six-faced die, the probability of getting a 4 is:

(A) certain
(B) 1 out of 2
(C) 1 out of 6
(D) 4 out of 6

52. One night the temperature rose from -4 degC to 9 degC. The rise in temperature was:

(A) 13 degC
(B) 5 degC
(C) -13 degC
(D) 9 degC

53. Write 5/(-8) with a positive denominator (its standard form).

(A) 5/8
(B) -8/5
(C) 8/5
(D) -5/8

54. A train moves at a uniform speed and covers 240 km in 4 hours. How far does it go in 1 hour?

(A) 240 km
(B) 60 km
(C) 4 km
(D) 120 km

55. An event that is certain to happen has a probability of:

- **(A)** 6
- **(B)** 0
- **(C)** 1/2
- **(D)** 1

56. A liquid cooled from 10 degC down to -6 degC. The fall in temperature was:

- **(A)** 6 degC
- **(B)** 4 degC
- **(C)** -16 degC
- **(D)** 16 degC

57. Which fraction is equal to 2/3?

- **(A)** 4/9
- **(B)** 6/8
- **(C)** 8/12
- **(D)** 3/2

58. Which of these is the FASTEST speed?

- **(A)** they are all equal
- **(B)** 10 m/s
- **(C)** 30 km/h
- **(D)** 1 km per minute

59. An event that can never happen has a probability of:

- **(A)** 10
- **(B)** 1/2
- **(C)** 0
- **(D)** 1

60. In cold weather, dark-coloured clothes can feel warmer than light ones because dark surfaces:

- **(A)** make heat by themselves
- **(B)** absorb more heat (radiation)
- **(C)** reflect more heat away
- **(D)** turn heat into magnetism

61. Among -1, -1/2, 0 and 1/2, the smallest number is:

- **(A)** -1/2
- **(B)** 1/2
- **(C)** -1
- **(D)** 0

62. A car's position is noted every second as 0 m, 5 m, 10 m, 15 m. The car's speed is:

- **(A)** 5 m/s
- **(B)** 10 m/s
- **(C)** 0 m/s
- **(D)** 15 m/s

63. Information collected as numbers or facts for a definite purpose is called:

(A) a graph
(B) data
(C) an average
(D) a range

64. Light-coloured clothes feel cooler in summer because they:

(A) create a cooling breeze
(B) absorb most of the heat
(C) reflect most of the heat away
(D) freeze the surrounding air

65. The number 0 can be written as a rational number in the form:

(A) 5/0
(B) 0 cannot be rational
(C) 0/5
(D) 1/0

66. If the length of a simple pendulum is increased, its time period:

(A) becomes zero
(B) increases
(C) decreases
(D) stays exactly the same

67. To decide the single most popular flavour of ice cream sold, the best average to use is the:

(A) mode
(B) median
(C) mean
(D) range

68. A laboratory thermometer differs from a clinical one mainly because the laboratory one:

(A) has a wider temperature range and no kink to hold the reading
(B) is always shorter than a clinical one
(C) can only measure body temperature
(D) has no scale of numbers

69. Which of these equals the whole number -2?

(A) -6/2
(B) -3/6
(C) -6/3
(D) -2/3

70. An object that is not moving has a speed of:

(A) its top speed
(B) equal to the time
(C) zero
(D) one

71. If every value in a data set is 8, then the mean of the set is:

(A) 0
(B) 16
(C) 8
(D) 4

72. The small kink (bend) in a clinical thermometer is there to:

(A) stop the mercury from slipping back so you can read the value
(B) let the mercury fall back instantly
(C) make the thermometer look attractive
(D) measure two temperatures at once

73. Between any two different rational numbers there are:

(A) only ten
(B) infinitely many rational numbers
(C) exactly one
(D) none at all

74. Two stations are 180 km apart and a train takes 3 hours between them. Its average speed is:

(A) 183 km/h
(B) 60 km/h
(C) 30 km/h
(D) 540 km/h

75. On five nights the temperatures were -2, 0, -3, 1 and -1 (in degC). The lowest temperature was:

(A) 0 degC
(B) 1 degC
(C) -3 degC
(D) -1 degC

76. A patient's temperature readings were 38, 39, 40 and 39 degC. The mean (average) temperature was:

(A) 156 degC
(B) 38 degC
(C) 39 degC
(D) 40 degC

77. Find (-3/4) x 8.

(A) -6
(B) -3/32
(C) -24
(D) 6

78. The to-and-fro motion of a child on a swing is an example of:

- **(A)** straight-line motion
- **(B)** no motion at all
- **(C)** periodic (oscillatory) motion
- **(D)** random motion

79. On those same five nights (-2, 0, -3, 1, -1 degC), the range of temperature was:

- **(A)** -2 degC
- **(B)** 3 degC
- **(C)** 2 degC
- **(D)** 4 degC

80. Why can two thin blankets keep you warmer than a single thick blanket of the same total thickness?

- **(A)** two blankets weigh less
- **(B)** air conducts heat extremely well
- **(C)** the blankets generate heat between them
- **(D)** the air trapped between the two blankets is a poor conductor of heat

81. The rational number 3/4 written as a decimal is:

- **(A)** 0.75
- **(B)** 0.43
- **(C)** 0.34
- **(D)** 1.33

82. Which relation correctly gives the distance travelled?

- **(A)** distance = speed / time
- **(B)** speed = distance x time
- **(C)** distance = speed x time
- **(D)** time = speed x distance

83. A bar graph is most suitable for:

- **(A)** comparing the quantities of different items
- **(B)** hiding small values
- **(C)** showing how something changes every second
- **(D)** listing names with no numbers

84. Compared with the sea, land usually heats up and cools down:

- **(A)** at exactly the same rate
- **(B)** faster
- **(C)** not at all
- **(D)** slower

85. Which of these rational numbers is the largest?

- **(A)** -3/4
- **(B)** -1
- **(C)** -1/2
- **(D)** -1/4

86. A car travels at a steady 40 km/h for 3 hours. The distance it covers is:

- **(A)** 43 km
- **(B)** 120 km
- **(C)** 13 km
- **(D)** 400 km

87. Find the mean of the first five whole numbers 0, 1, 2, 3 and 4.

- **(A)** 2
- **(B)** 10
- **(C)** 2.5
- **(D)** 3

88. A room heater warms the air near it; this warm air rises and cooler air moves in to replace it. This sets up:

- **(A)** an electric current
- **(B)** conduction through the walls
- **(C)** radiation only
- **(D)** convection currents

89. What must be added to -2/5 to get 0?

- **(A)** 0
- **(B)** 5/2
- **(C)** 2/5
- **(D)** -2/5

90. Sand clocks, sundials and water clocks were all early devices used to measure:

- **(A)** temperature
- **(B)** distance
- **(C)** speed
- **(D)** time

91. For the data 7, 7, 7, 8 and 9, the mean, median and mode are respectively:

- **(A)** 7.6, 8 and 7
- **(B)** 7, 7.6 and 8
- **(C)** 7.6, 7 and 7
- **(D)** 7, 7 and 7

92. At sea level, water boils at a temperature of about:

(A) 0 degC
(B) 50 degC
(C) 100 degC
(D) 37 degC

93. On the number line, the rational number -1/2 is located:

(A) halfway between 0 and 1
(B) halfway between -1 and 0
(C) exactly at -1
(D) exactly at 0

94. If a body covers a greater distance than another in the same time, then its speed is:

(A) greater
(B) exactly the same
(C) zero
(D) smaller

95. A car's speed at five checkpoints was 40, 50, 60, 50 and 50 km/h. The mode of these speeds is:

(A) 60 km/h
(B) 50 km/h
(C) 50 km/h is wrong, the mean is the mode
(D) 40 km/h

96. At normal pressure, ice melts (and water freezes) at a temperature of about:

(A) -10 degC
(B) 37 degC
(C) 0 degC
(D) 100 degC

97. Which of these two rational numbers is the smaller: -5/6 or -1/6?

(A) -1/6
(B) -5/6
(C) neither, both are positive
(D) they are equal

98. A cyclist covers 100 m in 4 s and then the next 100 m in 5 s. This shows the cyclist's motion is:

(A) uniform (the speed is constant)
(B) circular motion
(C) non-uniform (the speed is changing)
(D) no motion at all

99. Numbers or observations collected before they are organised in any way are called:

(A) an average
(B) a frequency table
(C) a bar graph
(D) raw data

100. At night a land breeze blows from the land towards the sea. This happens because at night the:

- **(A)** sea freezes every night
- **(B)** land stays hotter than in the daytime
- **(C)** sea cools down faster than the land
- **(D)** land cools down faster than the sea